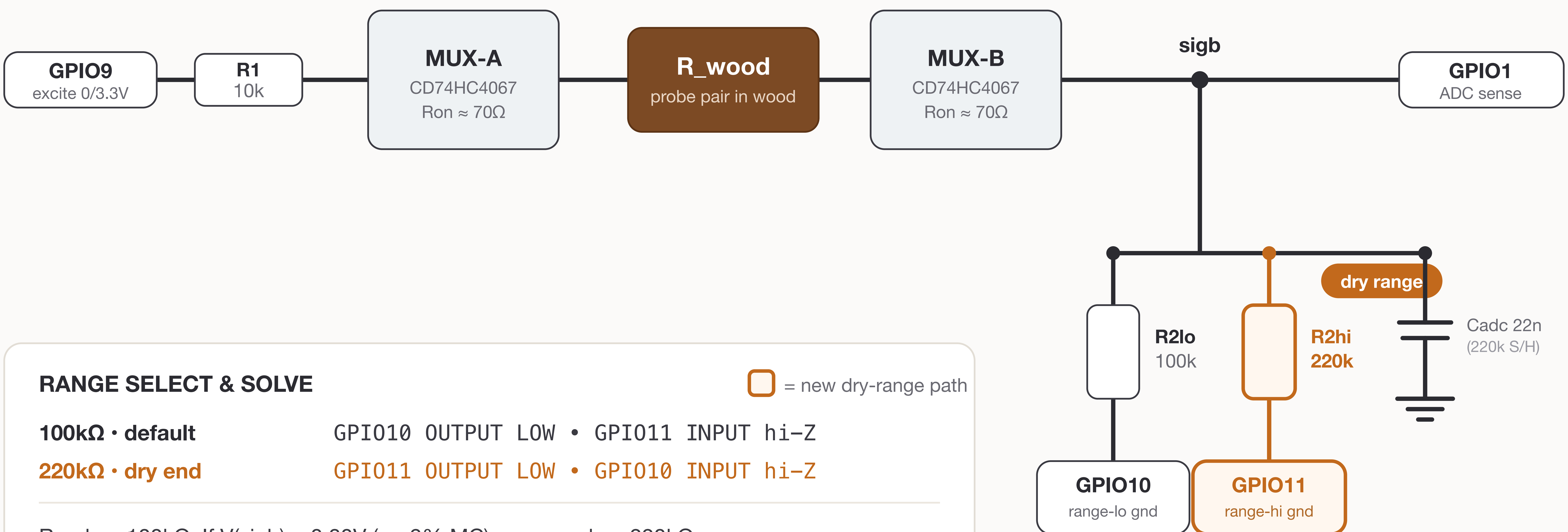
Wood-Moisture AFE — Switchable-R2 Auto-Range

One probe channel. A 220kΩ dry-range pull-down sits beside the stock 100kΩ; firmware picks the range per read.



RANGE SELECT & SOLVE

= new dry-range path

100kΩ · default	`GPIO10 OUTPUT LOW • GPIO11 INPUT hi-Z`
220kΩ · dry end	`GPIO11 OUTPUT LOW • GPIO10 INPUT hi-Z`

Read on 100kΩ. If V(sigb) < 0.80V (≤ ~9% MC) → re-read on 220kΩ.

`R_wood = R2 · VCC / V − R1 − R2 (use active R2)`

220kΩ lifts 6% MC from **0.205V** → **0.42V**, off the ADC noise floor.

MUX address lines (S0–S3 = GPIO4–7), EN = GPIO8, shared across MUX-A/B (not shown).